



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
MONDAY TEST

CLASS: X
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 21/07/25

General Instructions:

- This paper consists of three printed pages.
- All questions are compulsory.
- Marks will be deducted for spelling errors.

SECTION A
(Attempt all questions.)

Question 1

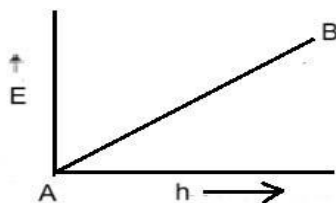
Choose the correct answers to the questions from the given options.

[1 × 8 = 8]

(i) Why are buses and trucks more likely to tip over, on sharp turns compared to cars?

- (a) They have wider tires
- (b) They have a raised center of gravity
- (c) They have a lowered center of gravity
- (d) Leaning causes a shift in the centre of gravity.

(ii) In the given graph of energy E versus height h (increasing from ground), line AB represents:



- (a) kinetic energy (b) potential energy (c) total energy (d) total mechanical energy

(iii) Assertion (A): In ideal condition, work done by a mass tied at the end of a string, rotating uniformly in a circle is zero.

Reason (R): Work done is zero if only the displacement is zero.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true and R is not the correct explanation of A.
- (c) A is false but R is true.
- (d) A is true but R is false.

(iv) If the vertical line through an object's centre of gravity falls outside its base, what is likely to happen?

- (a) The object becomes more stable.
- (b) The object stays in equilibrium.
- (c) The object topples over.
- (d) The weight of the object becomes zero.

(v) In which of the following, while in use, energy changes from chemical to electrical?

- (a) Loudspeaker (b) Burning coal (c) Electric cell (d) A steam engine

(vi) A body is acted upon by two unequal forces in opposite directions, but not in the same line. The effect is that the body will have:

- (a) only the rotational motion
(b) only the translational motion
(c) neither the rotational motion nor the translational motion
(d) rotational as well as translational motion.

(vii) The energy gained by an electron when it is accelerated through a potential difference of 10 volt is

- (a) 1.6×10^{-19} erg (b) 1.6×10^{-18} erg (c) 1.6×10^{-7} erg (d) 1.6×10^{-11} erg

(viii) An object is falling from a height of 10 m. When it falls by 3 m, which of the given statements is correct?

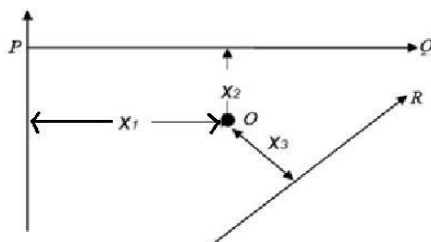
- (a) Potential energy is greater than kinetic energy.
(b) Kinetic energy is greater than potential energy.
(c) Both forms of mechanical energy are equal in magnitude.
(d) Total mechanical energy increases.

Question 2

(i) Complete the following by choosing the correct answers from the bracket: [3]

- (a) When a body is in the state of rest under the influence of several forces, the body is in _____ [static/translational /rotational] equilibrium.
(b) In the presence of friction and air resistance, the total mechanical energy of a freely falling object _____ [increases/decreases/remains same]
(c) The work done by the buoyant force, when a coin sinks, is _____ [positive/negative/zero].

(ii) P, Q and R are three coplanar forces that act at a distance x_1 , x_2 and x_3 respectively from a point O as shown in the figure. The point O is located on the same plane as that of the forces. Given that $P = Q = R = 6\text{N}$ and $x_1 = 4\text{m}$, $x_2 = 3\text{m}$ and $x_3 = 1\text{m}$. Based on this, find the net moment of force about O. [2]



Question 3

(i) A catapult throws a stone of mass 0.20 kg with a velocity 30 ms^{-1} . If 20% of the P.E. of the elastic band is wasted during transmission, find the magnitude of the potential energy. [2]

(ii) (a) Explain why a person running upstairs uses more power than a person walking upstairs.

- (b) How does the position of centre of gravity of a sphere change when it is completely filled with water?
[2]

(iii) Uniform circular motion refers to the movement of an object along a circular path at a constant speed. Throughout this motion, the object's distance from the axis of rotation remains unchanged. A force known as centripetal force, acting perpendicular to the object's motion is necessary to maintain this circular path by continuously altering the object's direction. On the other hand, centrifugal force is a fictitious force that appears to act on an object in non-inertial reference frames.

- (a) How is a uniform circular motion different from a non-uniform circular motion?
(b) What is the direction of centrifugal force?
(c) Is the statement true or false? "In uniform circular motion, the object's acceleration remains constant."

[3]

SECTION B

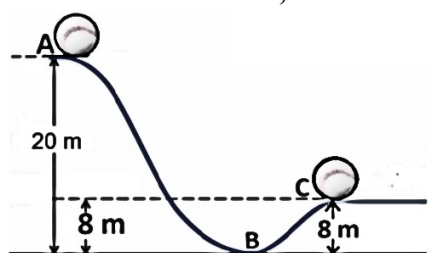
(Attempt all questions.)

Question 4

[3+3+4 =

10]

- (i) (a) A ball of mass 15 g is thrown vertically upward with initial velocity 20 ms^{-1} . At what height above the ground the ball would have equal potential and kinetic energy?
[Take $g = 10 \text{ ms}^{-2}$].
(b) Why is the stone of hand flour grinder provided with a handle near its rim?
- (ii) (a) Name the physical quantity expressed in Horse Power.
(b) If the kinetic energy of a body becomes twice its original value, what will be the new linear momentum in terms of the original linear momentum?
- (iii) A smooth sphere of mass 8 kg as shown in the diagram rolls on a smooth curved surface from the point A with a speed of 15 ms^{-1} . The sphere reaches point C passing through the point B. Based on the information, answer the following questions. [$g = 10 \text{ ms}^{-2}$].



- (a) Find the total mechanical energy of the sphere at the point A.
(b) Find the kinetic energy at the point B.
(c) Calculate the kinetic energy at C.

Question 5

[3+3+4 =

10]

- (i) A uniform meter rule is pivoted at 60 cm mark. The ruler is kept balanced when a load of 36 gf is hung at 90 cm mark.
(a) Draw the diagram for the arrangement.
(b) Calculate the weight of the meter rule.
- (ii) (a) Why are the railway wagons not loaded beyond a certain limit?
(b) State the conditions for equilibrium.
- (iii) A car of mass 120 kg is moving at a speed of 18 kmh^{-1} and it accelerates to attain a speed of 54 kmh^{-1} in 5 seconds.

- (a) Use the work-energy theorem to find the amount of work performed by the engine.**
- (b) Evaluate the power generated by the engine.**
- (c) What can be inferred about the sign of work done if a body's kinetic energy decreases?**